

Om Patel

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EDUCATION

California State University, Long Beach
Bachelor of Science in Computer Science

Jan 2024 – Dec 2027
Long Beach, CA

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

California State University, Long Beach

Long Beach, CA

- Developing an AI agent on a **C#/.NET** backend integrated with **ServiceNow** that classifies, triages, and auto-resolves incident and service-request tickets, on track to cut organization-wide ticket volume by **30%**.
- Delivered real-time **ServiceNow REST API** dashboards surfacing ticket trends, recurring failure modes, and high-volume categories, equipping **5** support agents to prioritize automation targets and reclaim **10 hrs/week** of manual triage.
- Designed **confidence-based routing** that resolves tickets only on strong knowledge-base retrieval and escalates ambiguous cases to human agents, scaling automation while preserving resolution accuracy.
- Implemented multithreaded **Java/C++** programs to study **parallel execution**, tuning task-scheduling and workload-partitioning to accelerate CPU-bound workloads across thread pools.
- Partnered with faculty and fellow researchers through weekly reviews, pair-debugging sessions, and shared benchmarking to keep experiments reproducible across the team.

Software Engineer Intern

July 2025 – June 2026

Beach Building Services (CSULB)

Long Beach, CA

- Integrated a **Gemini LLM** feature into a live web application, using prompt pipelines, input validation, and rate-aware fallbacks to sustain **100+ requests/min** with **zero downtime**.
- Modeled a normalized **PostgreSQL** schema with analytics views, indexed lookups, and aggregation queries, powering stakeholder reporting across professor and course rating data.
- Optimized performance-critical **C/C++** parsing and validation utilities on **Linux**, reducing processing time **30%** and unlocking higher data throughput for downstream reporting via memory-efficient algorithms and low-overhead I/O.
- Collaborated across a **cross-functional Scrum team** through sprint planning, **code reviews**, and daily stand-ups, shipping features on a biweekly release cadence.

AI Engineer Intern

September 2025 – December 2025

Impacter AI

San Francisco, CA

- Deployed **Python REST API** microservices with **ETL** data-transformation pipelines that embedded LLM inference into production, trimming end-to-end latency by **180ms** through async processing and response caching.
- Resolved concurrent-request **race conditions** and lifted LLM output accuracy by **12%** through output schemas, post-processing validation, and prompt engineering.
- Fine-tuned **GPT-3** and **Llama 2** with few-shot prompting and quantization, cutting compute costs by **35%** and average token usage by **28%** across workloads.
- Coordinated with backend and ML engineers via **pull-request reviews** and design docs, mentoring a peer intern on prompt-engineering practices.

PROJECTS

RepoSage AI | *Python, FastAPI, LangChain, PostgreSQL, pgvector, OpenAI API, GitHub API, Redis, BullMQ, Docker*

May 2026

- Engineered an **agentic RAG** system over GitHub repositories with **LangChain** tool-calling, enabling multi-step code understanding, architecture explanation, and bug detection across **50k+ line** codebases.
- Orchestrated an asynchronous distributed embedding pipeline with **BullMQ** and **pgvector**, applying semantic chunking to improve retrieval precision by **38%** over naive fixed-size approaches.

SimmSilos | *Python, FastAPI, TypeScript, PostgreSQL, Redis, Docker, Apache*

Mar 2026

- Created a **TypeScript VS Code extension** implementing a custom **virtual filesystem** with full save support, letting developers browse, edit, and write files inside isolated silos alongside live task and branch sidebars.
- Provisioned ephemeral developer environments: a **port registry** assigns each developer a dedicated **Docker container** on a unique port; on branch submission the container tears down, logs persist for audit, and the port frees for the next developer.

DevSecOps Project | *Jenkins, Docker, Kubernetes, Helm, ArgoCD, SonarQube, Trivy, Prometheus, Grafana, AWS*

Feb 2025

- Built a **Jenkins pipeline** with **Docker build/scan (SonarQube, Trivy)**, blocking **15+ vulnerabilities** before deployment.
- Automated deployments with **Helm + ArgoCD** on **Kubernetes**, streamlining **GitOps workflows** and improving delivery speed.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, Go, Java, JavaScript, TypeScript

AI/ML & Data: PyTorch, LangChain, RAG, Embeddings, Agents, Fine-tuning, Prompt Engineering, Airflow, dbt, Kafka, Spark, ETL/ELT, pandas, NumPy, OpenCV, MLflow

Backend & Storage: FastAPI, Node.js, Flask, PostgreSQL, pgvector, BigQuery, MySQL, MongoDB, Redis, AWS, Google Cloud, Firebase

Systems & Tools: Linux, Git, Docker, Kubernetes, CI/CD, GitHub Actions, REST APIs, WebSockets, Prometheus, Grafana, LangSmith, Agile/Scrum